

Nabh Garg

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B.Tech Computer Science Engineering student with strong Python fundamentals, hands-on backend development experience (Flask, FastAPI, REST APIs), and a solid Data Structures & Algorithms foundation – seeking to build production-grade backend systems as a Backend Intern.

Education

VIT Bhopal University

Sept 2023 – May 2027

B.Tech, Computer Science Engineering – CGPA: 8.2, No Active Backlogs

DAV Kalisindh Thermal Public School

2021 – 2023

Senior Secondary (Class XII) – 70%+

Technical Skills

Languages: Python, Java, C++, JavaScript

Backend & APIs: Flask, FastAPI, REST API Design, Node.js, Express.js, OOP

Databases: MySQL (Relational DB Design), MongoDB

Data Structures & Algorithms: Consistent competitive programming practice (Java, Python) – arrays, graphs, trees, DP, greedy

AI Tooling & Agents: Claude, Gemini API, LLM-based agentic pipelines, prompt engineering

Cloud & DevOps: AWS (Cloud Practitioner), Git/GitHub, Vercel, Render, Streamlit Cloud

Security: Web App Pentesting, Access Control Fundamentals, Burp Suite, CVSS v3.1

Core CS: OOPs, DBMS, Operating Systems, Computer Networks

Experience

Dholpur Combined Cycle Power Plant (RVUNL)

May 2026 – Jul 2026

Semester Internship – Computer Science Focus (45 Days, On-Site)

- Studied the GE Mark VIe Distributed Control System (DCS) architecture governing gas turbine operations, including startup sequencing logic, protective trip design, and real-time parameter control.
- Analyzed SCADA principles, industrial HMI design, and plant communication protocols (Modbus, EtherNet/IP, OPC DA/UA) underlying real-time, fault-tolerant industrial systems.
- Documented findings connecting large-scale industrial control systems to core CS concepts – real-time systems, fault tolerance, and industrial networking.

Projects

ELLE – Environmental Lens for Liquid Evaluation

[Live](#) | [GitHub](#)

- Built the backend (Python, Flask) for a full-stack AI platform that automates microplastic detection from images via a single REST API call with 90.9% accuracy, replacing costly lab equipment.
- Debugged and resolved a production out-of-memory crash on a 512MB RAM cloud instance by re-architecting the YOLOv8 inference pipeline from PyTorch to an ONNX-headless design with lazy model loading – a hard constraint-driven optimization problem.
- Designed MongoDB aggregation pipelines and schema for real-time analytics, plus a 300-second cache invalidation layer to keep read latency low under load.
- Built an IoT hardware bridge (Node.js, MQTT) for real-time sensor telemetry ingestion, cutting CPU usage by 70%.

SehPaathi – AI Exam & Document Assistant

[Live](#) | [GitHub](#)

- Engineered a Python backend service handling a document-analysis pipeline via the Gemini Files API, with automatic state polling for interactive Q&A over uploaded PDFs and external URLs.
- Architected a secure, multi-module system with persistent session-state management isolated across three independent tools, and environment-secret-based API key handling.
- Built and deployed backend logic for intent detection, multi-turn chat, and a dynamic day-wise study planner.

Certifications

- AWS Certified Cloud Practitioner** (CLF-C02) – Practice-track, Sept 2025
- The Bits and Bytes of Computer Networking** – Coursera (Google), Nov 2024 – [Verify Credential](#)
- Graphs Camp** Completion Certificate – [Verify Credential](#)

Achievements

- 1st Place** – CodeVerse Hackathon 2025 (200+ teams), VIT Bhopal – built a full-stack IoT health platform
- 3rd Place** – Blockchain Nexus, Techkriti'25, IIT Kanpur
- Galactic Problem Solver** – NASA International Space Apps Challenge

Leadership

Mharo Rajasthan Club, VIT Bhopal University

May 2025 – Present

Technical Team Lead

- Led technical operations for one of the university's largest student organizations, overseeing 1,000+ event registrations end-to-end and a flagship event with 1,500+ entries.

Unnat Bharat Abhiyan (Ministry of Education), VIT Bhopal

Nov 2025 – Present

Innovation & Technology Coordinator

- Develop technology-driven solutions addressing real-world rural community challenges under a Government of India initiative.

Internshala

Jun 2026 – Present

Student Partner – VIT Bhopal

- Represent the platform on campus, driving student engagement with training programs and internship opportunities.